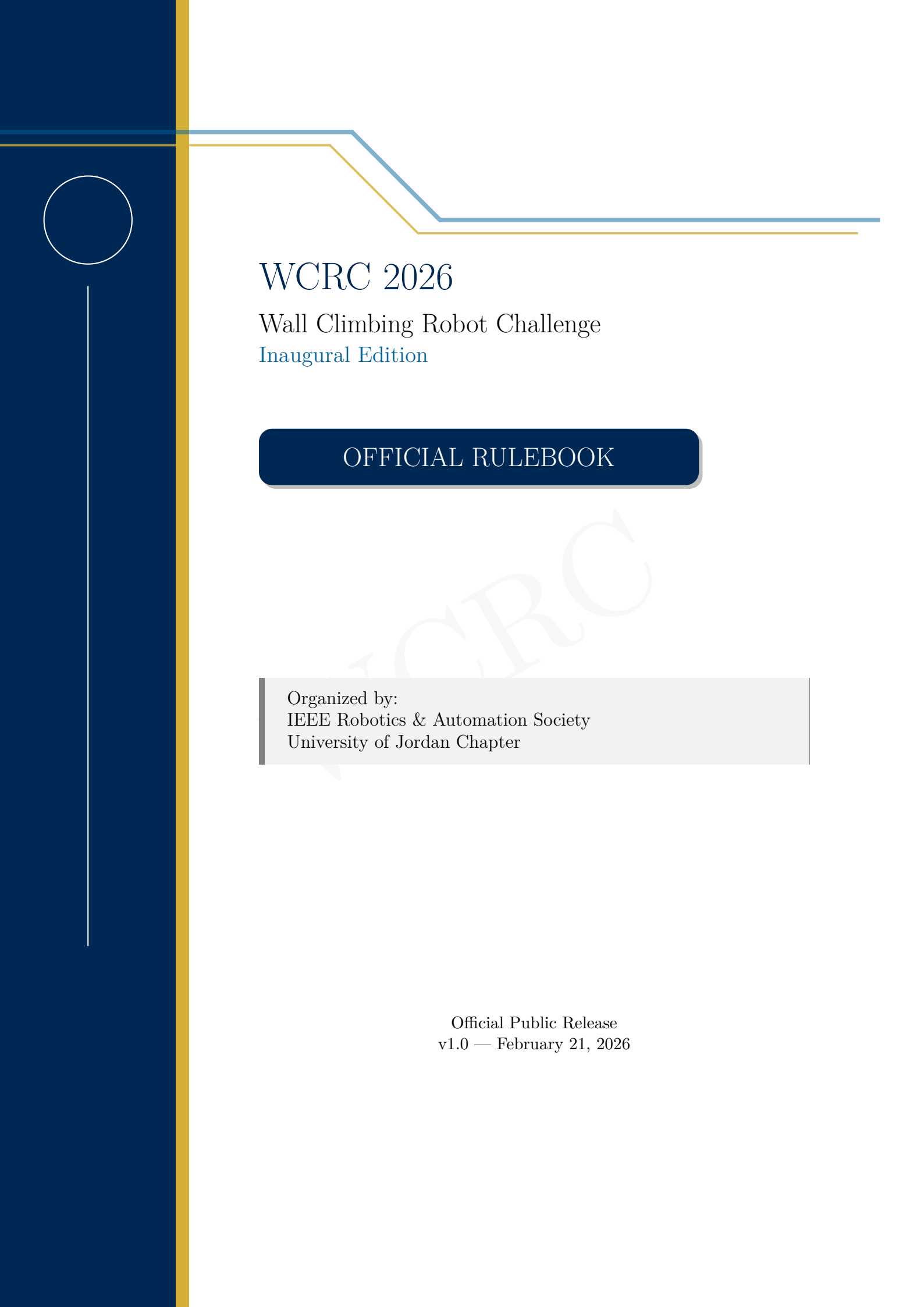




WCRC 2026

Wall Climbing Robot Challenge
Inaugural Edition

OFFICIAL RULEBOOK

Organized by:
IEEE Robotics & Automation Society
University of Jordan Chapter

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1 General Information

Overview, Venue & Dates

The Wall Climbing Robot Challenge (WCRC) aims to bridge the gap between mechanical design and robotics. Teams are tasked with engineering a robot capable of climbing vertical surfaces in the shortest possible time.

- Competition Dates: July 2 – 4, 2026.
- Venue: Fikr AI Hub & Study House.
- Eligibility: Open to all (University students, School students, and Hobbyists).

Team Composition Rules

- Team Size: Each team may consist of a maximum of 4 Members.
- Leadership: Each team must designate One Team Leader who will serve as the sole point of contact with referees and officials.
- One Robot Policy: Each registered team is permitted to compete with exactly ONE robot. Entering multiple robots under the same team name or interchanging robots during the competition is strictly prohibited.

2 The Arena Specifications

Engineering Warnings: Surface Materials

Teams must carefully design for the specific properties of the competition wall:

- Magnetic Wall: Constructed from 2.0 mm Galvanized Steel. Teams must account for Magnetic Saturation to ensure sufficient adhesion.

Figure 1: Official Arena Blueprint



- Competition Wall (Magnetic): Galvanized Steel (2.0 mm thick).
- Surface Condition: The wall will be wiped and cleaned between every single run to ensure consistent and fair adhesion conditions for all competitors.

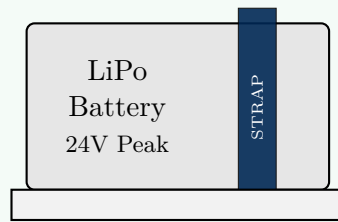
Arena Dimensions: The wall is 240 cm tall $\times$ 120 cm wide. The finish line is located at 200 cm from the ground. The 40 cm zone above the finish line serves as a safety buffer and overflow zone. The robot must remain entirely within the 120 cm wall width during the climb. Any part of the robot contacting or extending beyond the lateral edges of the wall results in DNF.

3 Robot Specifications

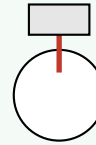
General Constraints

- Control Method: Hybrid (Autonomous OR RC). Fastest robot wins regardless of type.
- Weight: Max 5.0 kg.
- Start Size & Expansion: Max $30 \times 30 \times 30$ cm initially. Post-launch horizontal expansion is unlimited, provided the robot remains strictly on the face of the wall. Any contact with wall edges results in DNF.
- Main Chassis (Strict Definition): Defined as the structural frame housing the Microcontroller, Battery, and ALL actuators/motors responsible for the robot's upward propulsion.
- Cohesion: The robot must climb as a single cohesive unit; separation of the chassis from the climbing mechanism is prohibited. Note: Telescopic arms, extendable probes, and articulated mechanisms are permitted as long as they remain mechanically attached to the main chassis throughout the entire climb.

Power System & Battery Safety Standards



(A) Secure Mounting



(B) Master Switch Accessible

1. Allowed Battery Chemistries: LiPo, Li-ion, LiFePO₄, NiMH. (Lead-Acid BANNED).

2. Voltage & Master Switch:

- Max Voltage: 24V Peak Voltage. This limit applies to the maximum voltage measured at any point in the circuit, including charged capacitors. Nominal battery voltage must not exceed 22.2V (6S LiPo).
- Master Switch: A mechanism to completely disconnect power is required.
- E-Stop Functionality: The E-Stop must physically CUT electrical power to 0V. Using the E-Stop mechanism to discharge capacitors or trigger a speed boost is **STRICTLY PROHIBITED**.
- Insulation: Exposed copper or "twist-and-tape" connections are BANNED.

3. Mounting & Charging:

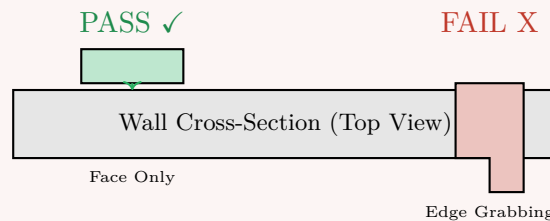
- Secure Mounting: Batteries must be fixed using Solid Holders OR Battery Straps. Battery must remain rigid.
- No In-Run Charging: Charging batteries during the run is prohibited.

PROHIBITED MECHANISMS

To ensure fair play and safety, the following are **STRICTLY BANNED**:

1. External Influence: Wireless power transfer, external magnetic fields, or remote processing (off-board brains) are **BANNED**.
2. Liquids & Chemical Adhesion: Glue, sticky tape, water, gels, or any substance that leaves residue (wet or dry).
3. Dropping Parts: Any part detaching from the robot during the run results in an immediate DNF. Intent is not considered. Note: “Minor rubber wear” is defined as passive residue (e.g., rubber dust, tread marks) left on the wall surface. Any discrete, solid piece visibly detaching from the robot during the run—regardless of size—constitutes a dropped part.
4. High-Pressure Systems: Compressed gas tanks (e.g., CO₂, HPA) are **BANNED** due to explosion risk. Vacuum pumps and fans (air movers) are **ALLOWED**.
5. Projectiles: Parts cannot be “shot” or detached to anchor. Extension must be mechanical.
6. Flying/Hovering: Drones are Disqualified.
7. Edge Grabbing: Climbing must be restricted to the face of the wall.
8. Destructive Tools: Spikes, drills, or sharp grippers.

Figure 2: No Edge Grabbing Rule



4 Competition Mechanics

4.1 Pre-Competition: Technical Inspection (Homologation)

Before Wave 1 begins, all robots must pass a mandatory technical inspection. Referees will explicitly verify:

- Maximum weight limit (5.0 kg) and Start Size (**30 × 30 × 30** cm).
- Proper installation and functionality of the Master Switch and E-Stop.
- Secure battery mounting.

Note: Robots failing to meet safety or dimensional constraints will not be permitted to compete.

4.2 Competition Structure: "Wave System"

The competition is divided into two waves:

- Starting Order: The run order for Wave 1 is determined by a Random Draw conducted during the Technical Inspection. Wave 2 follows the exact same sequence.
- Delay Policy: Teams must be in the staging area 2 runs before their turn. Missing the call results in a DNF for that specific run.
- Wave 1: All teams perform their First Run.
- Maintenance Break: A 30-minute compulsory break follows Wave 1.
 - Primary Climbing Mechanism (Definition): The complete system responsible for generating the robot's vertical movement, including: the motors and actuators, the transmission system (gears, belts, chains), and the adhesion method (magnets, vacuum, fans, or any other system). The robot in Wave 2 must climb using the same fundamental method as in Wave 1.
 - During the maintenance break, teams MAY:
 - ✓ Replace batteries
 - ✓ Replace wheels, tires, belts, or treads
 - ✓ Repair broken components (wires, sensors, mounts)
 - ✓ Adjust software or control settings
 - Teams MAY NOT:
 - ✗ Change the adhesion method (e.g., switching from magnets to vacuum fans)
 - ✗ Change the number, type, or arrangement of propulsion motors
 - ✗ Add or remove major structural components
 - ✗ Fundamentally change the robot's climbing strategy
- Wave 2: All teams perform their Second Run.

4.3 Winning Criteria: Fastest Best Run

The competition is a pure race against the clock. Measurements will be taken using high precision sensors. Important: The sensor provides a provisional time. The official time is confirmed only when the Head Referee validates (via VAR) that the Main Chassis (not an extension/arm) triggered the sensor and fully crossed the plane. All runs will be recorded from at least two angles for video review purposes.

Ranking System

Winner: The team with the Lowest Single Run Time (Best of 2).

Priority of Ranking (Tie-Breakers):

1. Fastest Time: (For robots that complete the full run).
2. Maximum Distance: (For robots that do not finish - DNF). In case of a tie in DNF distance, the winner is decided by: (a) Lowest Weight, (b) Shortest time to reach that specific distance.
3. Weight: (Lighter robot wins if time is tied).
4. Run Order: If all previous criteria are equal, the team with the earlier run number wins.

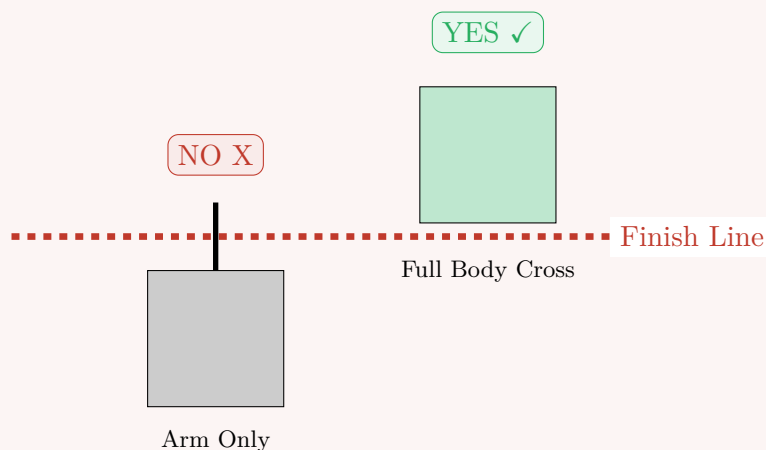
4.4 Run Procedures & Time Constraints

- Setup Time: Max 90 seconds.
- Start Zone Placement: The entire robot must be positioned completely within the 0-30 cm zone (as measured from the ground) before the start whistle. No part of the robot may extend beyond the 30 cm mark.
- Activation Procedure: Robot placed in Start Zone → Referee Whistle → Single interaction. For RC robots, this refers to the Start Command; continuous remote control is permitted thereafter.
- No Touch Policy: Once the robot is activated, teams must step back. Any further physical contact with the robot results in an immediate DNF.
- Max Run Time: 150 seconds per run.
- Fall Policy:
 - Zone A (0-30cm): A fall within this zone is a 'False Start' (1 restart allowed per run).
 - Zone B (>30cm): Any fall above the 30cm mark is an immediate DNF.
- Stall Policy: Any robot failing to achieve a minimum of 5 cm of upward progress within any 30 consecutive seconds, as judged by the referees, will be declared DNF. Minor vibrations or lateral movement without vertical gain do not constitute upward progress.

Finish Definition (Strict)

The run is complete ONLY when the ENTIRE MAIN CHASSIS (containing Battery & All Propulsion Motors) fully crosses the 200cm finish line plane. Lightweight arms/probes do not count.

Figure 3: Finish Line Crossing Rule



5 Safety Protocols

Safety Tethers & E-Stops

- Safety Tether (Optional): Passive safety rope allowed. Teams must provide their own tether line. The organizers will provide a standard anchor point/pulley at the top of the wall. Tethers must be non-elastic. It must remain visually slack at all times. The tether cannot be used for hoisting or providing any upward force. Any observed tension by referees/video review results in a DNF.
- Emergency Stop: Highly recommended (Physical or Wireless).
- Liability: Teams bear full responsibility for damage caused by falling.

6 Adjudication, Code of Conduct & Integrity

Ethics & Anti-Sabotage Rules

To ensure a fair and professional environment, the following rules are strictly enforced. Violations result in immediate Disqualification (DQ) and potential bans.

1. Anti-Sabotage: Deliberately leaving residue (oil, sticky fluids) or modifying the wall surface to hinder other competitors is strictly prohibited. (Note: Normal wear like minor rubber dust is exempt).
2. RF Interference & Controller Discipline: The use of signal jammers or malicious devices is strictly prohibited. Furthermore, to prevent accidental frequency overlap (e.g., 2.4GHz interference), all remote controllers and transmitters must remain POWERED OFF at all times in the Pit Area and audience seating. Teams are ONLY permitted to turn on their controllers when officially instructed by the referee inside the Start Zone. Violating this rule may result in immediate disqualification.
3. IP Respect (Pit Area): Teams must respect the intellectual property of others. Inspecting, touching, or photographing competitor robots in the pit area without explicit permission is forbidden.
4. Pit Access: Access to the Pit Area is restricted to registered team members wearing official badges.
5. Weighing Protocol:
 - Official Weighing: Occurs immediately BEFORE the robot enters the Start Zone for each run. The robot must enter the Start Zone in the exact configuration in which it was weighed. No additions or removals are permitted between the official weigh-in and the start of the run.
 - Spot Checks: Referees reserve the right to re-weigh any robot AFTER a run. If the post-run weight is more than 100g lighter than the pre-run weight, the run is voided (DQ).
6. Sportsmanship: Unsportsmanlike conduct, including verbal abuse of officials or competitors, will not be tolerated.
7. Language: The Official Rulebook is in English. Communication with judges is allowed in Arabic or English; however, the English text prevails in disputes.
8. Referee's Discretion & Appeals: In cases not explicitly covered, referees may make rulings based on the spirit of fair wall-climbing competition. Any official appeal to the Head Judge must be submitted by the Team Leader within 15 minutes of the disputed run. Appeals do not halt the competition; the next run will proceed while the Head Judge reviews the VAR. Late appeals will not be considered.
9. Media Rights: By participating in the WCRC, teams grant the IEEE RAS UJ Chapter the right to capture and use photographs and video recordings of their robots and team members for promotional, educational, and archival purposes.

7 Registration, Fees and Prizes

- Registration Deadline: June 15, 2026. (We advise early registration as spots are limited).
- Initial Inspection (Two Phases):
 - Phase 1 - Preliminary Review (Deadline: June 18, 2026): Teams must submit documentation (physical prototype photos or CAD design) via the official WCRC website. The organizing committee will respond within 3 business days with either approval or required modifications. This early review ensures the core concept complies with the rules and does not include any strictly banned mechanisms.
 - Phase 2 - On-Site Technical Inspection (July 2, 2026): The mandatory physical inspection conducted on competition day as described in Section 4.1. Note: Approval in Phase 1 does not guarantee passing the on-site Technical Inspection.
- Payment Method: Entry fees can be paid in cash to the chapter's Treasurer, Omar Eshtewe, or transferred via e-wallets (CliQ details will be provided on the official WCRC website).
- Entry Fees:
 - 5 JOD: Discounted rate if at least one team member holds an active IEEE membership.
 - 10 JOD: Standard rate for teams with no IEEE members.
- Safety Deposit: 10 JOD. (Refundable after the event. Total forfeiture occurs if any structural damage is caused to the arena surface or sensors).



Quick Reference Table

Parameter	Value
Max Weight	5.0 kg
Start Size	30 × 30 × 30 cm
Max Voltage	24V Peak
Wall Height	240 cm
Wall Width	120 cm
Finish Line	200 cm
Start Zone	0 – 30 cm
Setup Time	90 seconds
Max Run Time	150 seconds
Stall Timeout	30 seconds
Stall Minimum Progress	5 cm
False Start Zone	0 – 30 cm
Maintenance Break	30 minutes
Appeal Window	15 minutes
Post-Run Weight Tolerance	100g
Preliminary Review Deadline	June 18, 2026

Versioning Notice: This rulebook may be updated. Teams will be notified of any changes via the official WCRC website. The latest published version supersedes all previous versions.